SRINIVAS

Senior AI/ML Engineer | Generative AI Specialist
Bengaluru, India

PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with 8 years of experience spanning full-stack Python development, data science, NLP, and Generative AI. Adept at building end-to-end intelligent systems—from backend APIs and data pipelines to agentic AI workflows and autonomous web-crawling agents. Proven track record of delivering production-grade GenAI solutions including RAG systems, agentic automation, and self-learning crawling agents. Early adopter of OpenAI GPT-3 APIs, with deep hands-on expertise across the modern AI stack.

CORE COMPETENCIES

- Python / Django / FastAPI
- Generative AI / LLMs
- Agentic AI Systems
- NLP (NLTK, spaCy)
- RAG Architectures
- Deep Learning
- Selenium / Web Crawling
- Data Analytics & Science
- OpenAI / GPT APIs

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer

HCL Technologies | India 2023 – Present

- Architected and deployed RAG (Retrieval-Augmented Generation) pipelines, enabling accurate knowledge retrieval from large enterprise document repositories.
- Built agentic AI automation workflows for vendor invoice processing and sales order creation, significantly reducing manual turnaround time and error rates.
- Developed an intelligent chatbot that fetches real-time data on the fly and leverages GenAI to answer user queries with contextual, accurate responses.
- Currently leading development of a self-learning web crawling agent that accepts a target schema and URL, autonomously navigates websites to extract structured data, handles missing values with dynamic decision-making, and continuously improves performance by learning from previous crawl runs.
- Collaborated with cross-functional teams to integrate AI solutions into existing enterprise systems and client workflows.

Python Developer → Data Scientist → AI/ML Engineer

Infosight Consulting | India 2018 – 2023

- Started as a Python Backend Developer building RESTful APIs with Django; also developed Selenium-based web crawlers for automated data collection.
- Transitioned to Python backend development with Django and later FastAPI, building scalable, high-performance microservices and APIs.
- Moved into Data Analytics and Data Science, performing exploratory data analysis, building predictive models, and delivering actionable business insights using Python.
- Advanced into Deep Learning and NLP—built a vendor scoring system using NLTK and spaCy that analyzed vendor feedback, web presence, and other data sources to generate automated quality scores.
- Early adopter of OpenAI's GPT-3 API upon release; integrated GPT-3 Turbo into the vendor scoring project, transitioning the solution to a Generative AI–powered approach and marking a full pivot into GenAI.

EDUCATION

Bachelor of Engineering – Computer Science
Bapuji Institute of Engineering and Technology | Davangere, Karnataka

TECHNICAL SKILLS

Languages	Python
Frameworks	Django, FastAPI, Flask
AI/ML	OpenAI GPT APIs, LangChain, Deep Learning, NLP, RAG, Agentic AI
NLP Libraries	NLTK, spaCy, Transformers
Data Science	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Automation	Selenium, Custom Web Crawling Agents
Databases	PostgreSQL, MySQL, MongoDB, Vector Databases
Tools & Platforms	Git, Docker, Linux, REST APIs, CI/CD

KEY PROJECTS

Self-Learning Web Crawling Agent *(Current)*

- Autonomous agent that takes a schema definition and target URL, dynamically navigates websites, extracts structured data, and self-corrects for missing fields.
- Incorporates learning from previous runs to progressively reduce crawl time and improve extraction accuracy.

Intelligent Enterprise Chatbot

- GenAI-powered chatbot that retrieves real-time data and provides contextual, accurate answers to user queries across enterprise data sources.

Agentic Invoice & Sales Order Automation

- End-to-end agentic workflow that automates vendor invoice processing and sales order creation, reducing manual effort and error rates.

AI-Powered Vendor Scoring System

- NLP-driven scoring engine using NLTK, spaCy, and later GPT-3 to evaluate vendors based on feedback, web data, and other signals.